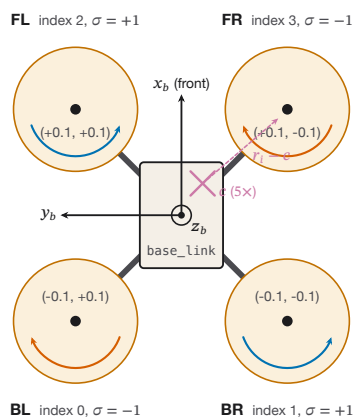


a Rotor layout order [back_left, back_right, front_left, front_right]



Top view, body axes (thrust axes $a_i = +z_b$, out of the page). Positions [m] are the ARL-Robot-1 USD rotor origins (the synthetic plant copies them). Arrows: reaction-torque sign σ_i of rotor_directions (+1 counter-clockwise, -1 clockwise about $+z_b$), the explicit simulation_diagonal_pairs choice, not a measured spin direction. Action order = index.

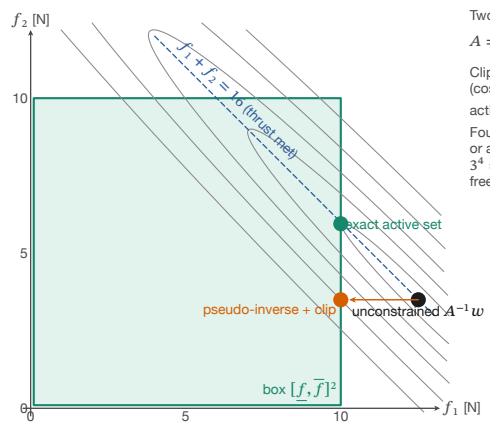
b Allocation matrix from actual geometry

	BL	BR	FL	FR	
F_x	0	0	0	0	F_x, F_y : no authority
F_y	0	0	0	0	
F_z	1	1	1	1	rank 4: f_z, M_x, M_y, M_z independent
M_x	+0.104	-0.096	+0.104	-0.096	
M_y	+0.106	+0.106	-0.094	-0.094	
M_z	-0.070	+0.070	+0.070	-0.070	

$$A_i = \left[(r_i - c) \times a_i + \sigma_i k_\tau a_i \right], \quad w = A f$$

Values: synthetic plant ($c = (6, -4, 2)$ mm, $k_\tau = 0.07$ m). Isaac builds A the same way from PhysX link poses and the resolved CoM; the upstream ± 0.13 m matrix is only logged as a difference. k_τ gives yaw far less leverage than the arms.

c Why not pseudo-inverse and clip BoundedAllocator



Two-rotor analogue (schematic): $w = (f_z, M) = (16 \text{ N}, 0.9 \text{ Nm})$, $A = \begin{bmatrix} 1 & 1 \\ 0.1 & -0.1 \end{bmatrix}$, $W = I$. Grey: level sets of $\|Af - w\|^2$. Clipping moves the solution off the thrust valley (cost 6.31); the exact optimum slides along the active bound $f_1 = \bar{f}$ (cost 0.25). Four rotors: every rotor is at its lower bound, free or at its upper bound, so the code enumerates $3^4 = 81$ active sets, solves least squares for the free rotors and keeps the cheapest feasible one.

d

